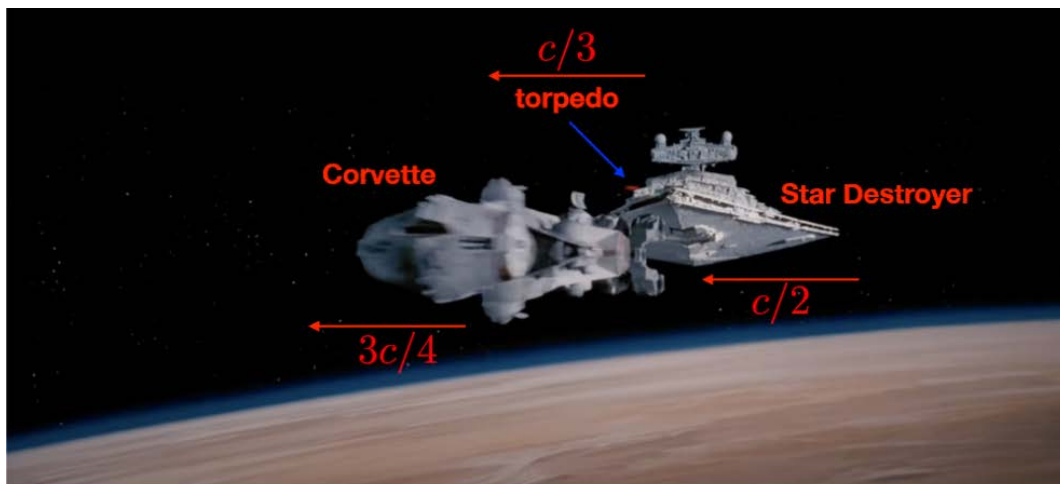


All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:  
Section:

1. (30 points) **Synchronizing clocks.** Suppose you have two identical clocks A and B located 1 m apart. Describe in detail a scheme to synchronize those clocks without moving the clocks themselves. You can assume (as in Discussion 1, Part 1) that you have access to any necessary materials: rulers, mirrors, lasers, stopwatches, additional people to help you, etc.
2. (30 points) **More practice with velocity addition.** Morin exercise 1.42 (p. 53)
3. (40 points) **Relative speeds in different frames.** You are on the planet Tatooine watching the opening scene of Star Wars Episode IV play out in the sky above you. An Imperial Star Destroyer (speed  $c/2$  with respect to you) is chasing Princess Leia's Corellian Corvette (speed  $3c/4$  with respect to you). The Star Destroyer fires a torpedo (with speed  $c/3$  in the frame of the Star Destroyer) toward the Corvette. See the diagram below. Does the ship escape?
  - (a) (20 points) First, compute the relative velocity between the Corvette and the torpedo, in *your* reference frame. If this relative velocity is positive, the ship escapes because the distance between it and the torpedo continues to increase.
  - (b) (20 points) Now, compute the relative velocity between the Corvette and the torpedo, in the frame of the Star Destroyer. You should find the same sign for the relative velocity as in part (a), consistent with the fact that all inertial observers must agree on whether or not the ship is hit by the torpedo, which would represent a single spacetime event.



Solutions:

1. There are many possible ways to do this. One way is to send a light signal between A and B and make sure that the time it takes light to go from A to B is the same as the time it takes the light to return from B to A. However, the speed of light in air is not the same as the speed of light in vacuum, so you should first build a vacuum tube between the two clocks. Place a mirror on the end of the tube attached to clock B, and connect a photosensor which detects the light that hits the mirror. Place a laser source at the location of clock A along with a photosensor. When you turn on the laser, clock A reads  $t_A$ . When the light from the laser reaches B, the photosensor clicks, telling you that clock B reads  $t_B$ . The light bounces off the mirror and returns to A, triggering the photosensor and telling you that clock A reads  $t_{A'}$ . The clocks are synchronized when  $t_B - t_A = t_{A'} - t_B$ ; in other words, adjust clock B such that  $t_B = \frac{1}{2}(t_A + t_{A'})$ . This is in fact the method that Einstein suggested in his 1905 paper!
2. Suppose  $C$  runs toward  $B$  with speed  $v$ . Then the relative velocity of  $B$  and  $C$ , as seen by  $C$ , is  $v$ , while the relative velocity of  $A$  and  $C$ , as seen by  $C$ , should also be  $v$ . But we are told that the relative velocity of  $A$  and  $B$ , as seen by  $B$ , is  $4c/5$ . So the relativistic addition of  $v$  and  $v$  should give  $4c/5$ . The rest is just algebra, which is made easier by letting  $\beta = v/c$ :

$$\frac{\beta + \beta}{1 + \beta^2} = \frac{4}{5} \implies 2\beta^2 - 5\beta + 2 = 0.$$

This quadratic equation has roots  $\beta = 1/2$  and  $\beta = 2$ , but the second root is spurious because it implies  $v = 2c$ , which is impossible. So the solution is  $\beta = 1/2$ , or  $v = c/2$ .

3. (a) In your frame, the torpedo travels at speed  $\frac{c/2+c/3}{1+(c/2)(c/3)/c^2} = \frac{5}{7}c$ . We are already given the speed of the Corvette in your frame, so the relative velocity of the torpedo and Corvette in your frame are given by subtracting the two speeds:  $v_{\text{rel,you}} = \frac{3}{4}c - \frac{5}{7}c = \frac{1}{28}c > 0$ . This is positive (the Corvette is going faster than the torpedo), so the ship escapes.
- (b) In the frame of the Star Destroyer, the torpedo speed is just  $c/3$ . The speed of the Corvette in the Star Destroyer frame can be found by using the velocity addition formula, as follows: the speed of the Corvette with respect to the planet is  $3c/4$ , and the speed of the planet with respect to the Star Destroyer is  $-c/2$ , so the speed of the Corvette with respect to the Star Destroyer is  $\frac{3c/4 - c/2}{1 + (3c/4)(-c/2)/c^2} = \frac{c/4}{5/8} = \frac{2}{5}c$ . The relative speed is again given by subtracting the speed of the torpedo from the speed of the Corvette, both measured in the Star Destroyer frame:  $v_{\text{rel,SD}} = \frac{2}{5}c - \frac{1}{3}c = \frac{1}{15}c > 0$ . This is still positive, so the ship still escapes in the Star Destroyer frame, as it must.